Experiment 2: To Perform Data Preprocessing Using Python

Student & Submission Details

Parameter	Details
Institute	Vivekanand Education Society's Institute of Technology (VESIT)
Department	Information Technology
Subject	Data Mining and Business Intelligence (DMBI)
Experiment Number	2
Experiment Title	To Perform Data Preprocessing Using Python
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Class	D15A
Dataset Topic	Food Delivery Management Dataset

Aim

To perform data preprocessing on a Food Delivery Management dataset using Python by handling missing values, removing duplicate records, encoding categorical data, and scaling numerical features.

Objectives

- Load and display a Food Delivery Management dataset using Python.
- Identify and handle missing values in both numerical and categorical attributes.
- Detect and remove duplicate records to maintain data integrity.
- Convert categorical attributes into numerical representations using label encoding.
- Scale numerical features using `StandardScaler` to normalize standard distributions.
- Verify and export the cleaned dataset as an Excel spreadsheet (`Preprocessed_Food_Delivery.xlsx`) for further analysis.

Software / Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas Library
- NumPy Library
- Scikit-learn (`sklearn.preprocessing`)
- Microsoft Excel / Compatible Spreadsheet Software

Theory

Data preprocessing is a pivotal phase in data mining and machine learning pipelines. Raw data collected from operational platforms—such as food delivery services—often contains noise, missing entries, duplicate transactions, and inconsistent formats. Processing raw data guarantees high data quality, preventing skewed models and inaccurate analytical insights.

The major preprocessing operations demonstrated in this experiment include:

1. **Missing-Value Handling**: Incomplete entries occur due to network dropped calls, user refusal, or missing records. Numerical missing values are imputed using column mean statistics to preserve sample mean behavior, while categorical missing values are imputed using mode (most frequent item).
2. **Duplicate Removal**: Duplicate rows skew analytical aggregates and introduce overfitting in predictive modeling. Repeated rows are identified via key attributes and dropped using Pandas `drop_duplicates()`.
3. **Categorical Encoding**: Machine learning algorithms require numerical values for matrix operations. `LabelEncoder` maps unique discrete text categories (such as `Food_Category`, `City`, and `Payment_Mode`) into ordinal integer levels ($0, 1, 2, \dots, n-1$).

4. **Feature Scaling**: Numerical attributes like order amounts, ratings, and delivery durations often possess distinct scales and units. `StandardScaler` standardizes features by centering the distribution around mean $\mu = 0$ with unit standard deviation $\sigma = 1$, using the standard transformation:
$$z = \frac{x - \mu}{\sigma}$$
5. **Final Verification and Export**: The resulting dataset is validated to confirm 0 missing values and 0 duplicate entries before being saved as an Excel file (`Preprocessed_Food_Delivery.xlsx`).

Dataset Description

The sample dataset represents order activity in a Food Delivery Management system. It initially consists of 10 records and 8 feature columns:

Column Name	Data Type	Description	Sample Value
Order_ID	Numerical (Int…	Unique identifier for each order transaction	1001
Customer	Categorical (T…	Name of the customer placing the order	Aarav
Food_Category	Categorical (T…	Type/Cuisine of food ordered	Pizza
City	Categorical (T…	Delivery location city	Mumbai
Delivery_Time	Numerical (Co…	Time taken to deliver order (minutes)	30
Order_Amount	Numerical (Co…	Total cost of order currency units	450
Rating	Numerical (Co…	Customer feedback score (1.0 - 5.0)	4.5
Payment_Mode	Categorical (T…	Method used for order payment	UPI

The raw dataset intentionally incorporates four missing values (across `Delivery_Time`, `Rating`, `Food_Category`, and `Payment_Mode`) and one exact duplicate entry (Order ID 1002). Upon successful preprocessing, the cleaned dataset is reduced to 9 unique records across 8 standardized columns.

Python Program

Python

```python
import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder, StandardScaler

# Create Food Delivery dataset
data = {
    "Order_ID": [1001, 1002, 1003, 1004, 1005, 1006, 1002, 1008, 1009, 1010],
    "Customer": ["Aarav", "Diya", "Kabir", "Meera", "Ishaan", "Anaya", "Diya",
"Rohan", "Sara", "Vivaan"],
    "Food_Category": ["Pizza", "Indian", "Chinese", "Burgers", None, "Indian",
"Indian", "Mexican", "Pizza", "Chinese"],
    "City": ["Mumbai", "Thane", "Mumbai", "Navi Mumbai", "Pune", "Thane", "Thane",
"Mumbai", "Pune", "Navi Mumbai"],
    "Delivery_Time": [30, 40, np.nan, 25, 35, 50, 40, 20, 45, 55],
    "Order_Amount": [450, 300, 550, 250, 400, 350, 300, 600, 500, 700],
    "Rating": [4.5, 4.0, 5.0, np.nan, 4.2, 3.8, 4.0, 4.7, 4.4, 4.1],
    "Payment_Mode": ["UPI", "Card", "Cash", "UPI", "Card", None, "Card", "UPI",
"Cash", "Card"]
}

df = pd.DataFrame(data)

# Display original dataset
print("===== ORIGINAL FOOD DELIVERY DATASET =====")
print(df)
print("\nDataset Shape:", df.shape)

# Check missing values
print("\n===== MISSING VALUES =====")
print(df.isnull().sum())
print("\nTotal Missing Values:", df.isnull().sum().sum())
```

```
# Handle missing values
df["Delivery_Time"] = df["Delivery_Time"].fillna(df["Delivery_Time"].mean())
df["Rating"] = df["Rating"].fillna(df["Rating"].mean())
df["Food_Category"] = df["Food_Category"].fillna(df["Food_Category"].mode()[0])
df["Payment_Mode"] = df["Payment_Mode"].fillna(df["Payment_Mode"].mode()[0])

print("\n===== DATASET AFTER HANDLING MISSING VALUES =====")
print(df)
print("\n===== MISSING VALUES AFTER HANDLING =====")
print(df.isnull().sum())

# Check and remove duplicates
print("\n===== DUPLICATE RECORDS =====")
print("Number of duplicate rows:", df.duplicated().sum())
print(df[df.duplicated()])

df = df.drop_duplicates()

print("\n===== DATASET AFTER REMOVING DUPLICATES =====")
print(df)
print("Duplicates after removal:", df.duplicated().sum())
print("Dataset Shape:", df.shape)

# Encode categorical columns
food_encoder = LabelEncoder()
city_encoder = LabelEncoder()
payment_encoder = LabelEncoder()

df["Food_Category"] = food_encoder.fit_transform(df["Food_Category"])
df["City"] = city_encoder.fit_transform(df["City"])
df["Payment_Mode"] = payment_encoder.fit_transform(df["Payment_Mode"])

print("\n===== DATASET AFTER CATEGORICAL ENCODING =====")
print(df)
print("\nFood Category Encoding:")
print(dict(zip(food_encoder.classes_,
food_encoder.transform(food_encoder.classes_))))
print("\nCity Encoding:")
print(dict(zip(city_encoder.classes_,
city_encoder.transform(city_encoder.classes_))))
print("\nPayment Mode Encoding:")
```

```
print(dict(zip(payment_encoder.classes_,
payment_encoder.transform(payment_encoder.classes_))))

# Feature scaling
numerical_columns = ["Delivery_Time", "Order_Amount", "Rating"]
scaler = StandardScaler()
df[numerical_columns] = scaler.fit_transform(df[numerical_columns])

print("\n===== DATASET AFTER FEATURE SCALING =====")
print(df)
print("\n===== SCALED NUMERICAL COLUMNS =====")
print(df[numerical_columns])

# Final verification
print("\n===== FINAL DATASET VERIFICATION =====")
print("\nMissing Values:")
print(df.isnull().sum())
print("\nTotal Missing Values:", df.isnull().sum().sum())
print("\nDuplicate Rows:")
print(df.duplicated().sum())
print("\nFinal Dataset Shape:", df.shape)
print("\nFinal Preprocessed Dataset:")
print(df)

# Export preprocessed dataset
output_file = "Preprocessed_Food_Delivery.xlsx"
df.to_excel(output_file, index=False)
print("\nPreprocessed dataset saved successfully!")
print("File name:", output_file)
```

Categorical Encoding Mappings

The categorical variables were successfully mapped to numeric values using `LabelEncoder` as follows:

Feature Name	Original Value	Encoded Numerical Value
Food_Category	Burgers	0

Feature Name	Original Value	Encoded Numerical Value
	Chinese	1
	Indian	2
	Mexican	3
	Pizza	4
City	Mumbai	0
	Navi Mumbai	1
	Pune	2
	Thane	3
Payment_Mode	Card	0
	Cash	1
	UPI	2

Expected Output and Screenshots

Execution Screenshots & Outputs

Figure 1: Importing required Python libraries (`pandas`, `numpy`, `sklearn.preprocessing`).

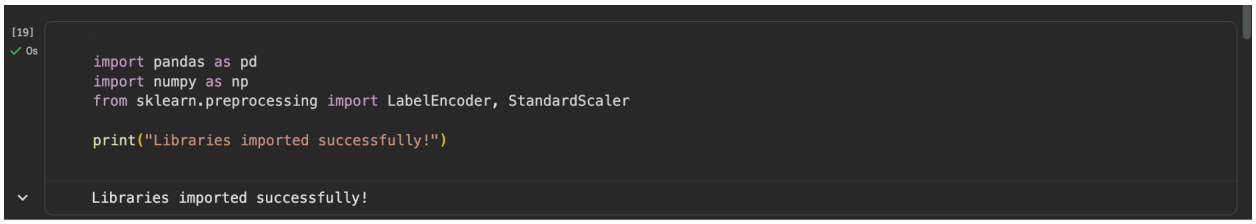


Figure 2: Defining and displaying the original Food Delivery Management dataset with shape (10, 8).

```
...  ===== ORIGINAL FOOD DELIVERY DATASET =====
    Order_ID Customer Food_Category         City  Delivery_Time  Order_Amount  \
0       1001    Aarav         Pizza       Mumbai           30.0           450
1       1002     Diya       Indian        Thane           40.0           300
2       1003    Kabir       Chinese       Mumbai            NaN           550
3       1004    Meera       Burgers  Navi Mumbai           25.0           250
4       1005   Ishaan          None         Pune           35.0           400
5       1006    Anaya       Indian        Thane           50.0           350
6       1002     Diya       Indian        Thane           40.0           300
7       1008    Rohan       Mexican       Mumbai           20.0           600
8       1009     Sara         Pizza         Pune           45.0           500
9       1010   Vivaan       Chinese  Navi Mumbai           55.0           700

    Rating Payment_Mode
0      4.5          UPI
1      4.0         Card
2      5.0         Cash
3      NaN          UPI
4      4.2         Card
5      3.8         None
6      4.0         Card
7      4.7          UPI
8      4.4         Cash
9      4.1         Card

Dataset Shape: (10, 8)
```

Figure 3: Identification of 4 missing values across Delivery_Time, Rating, Food_Category, and Payment_Mode.

```
...  ===== MISSING VALUES IN FOOD DELIVERY DATASET =====
    Order_ID          0
    Customer          0
    Food_Category     1
    City              0
    Delivery_Time     1
    Order_Amount      0
    Rating            1
    Payment_Mode      1
    dtype: int64

    Total Missing Values: 4
```

Figure 4: Handling missing values using mean/mode imputation and verifying zero missing remaining.

```
    ===== DATASET AFTER HANDLING MISSING VALUES =====
...    Order_ID Customer Food_Category         City  Delivery_Time  Order_Amount  \
    0      1001    Aarav         Pizza       Mumbai      30.000000           450
    1      1002     Diya        Indian        Thane      40.000000           300
    2      1003    Kabir       Chinese       Mumbai      37.777778           550
    3      1004    Meera       Burgers  Navi Mumbai      25.000000           250
    4      1005   Ishaan        Indian         Pune      35.000000           400
    5      1006    Anaya        Indian        Thane      50.000000           350
    6      1002     Diya        Indian        Thane      40.000000           300
    7      1008    Rohan       Mexican       Mumbai      20.000000           600
    8      1009     Sara         Pizza         Pune      45.000000           500
    9      1010   Vivaan       Chinese  Navi Mumbai      55.000000           700

       Rating Payment_Mode
    0     4.5          UPI
    1     4.0         Card
    2     5.0         Cash
    3     4.3          UPI
    4     4.2         Card
    5     3.8         Card
    6     4.0         Card
    7     4.7          UPI
    8     4.4         Cash
    9     4.1         Card

    ===== MISSING VALUES AFTER HANDLING =====
    Order_ID         0
    Customer         0
    Food_Category    0
    City             0
    Delivery_Time    0
    Order_Amount     0
    Rating           0
    Payment_Mode     0
    dtype: int64
```

Figure 5: Detection of 1 duplicate record in the dataset (Order_ID 1002).

```
...  ===== DATASET AFTER REMOVING DUPLICATES =====
       Order_ID Customer Food_Category         City  Delivery_Time  Order_Amount  \
    0      1001    Aarav         Pizza       Mumbai      30.000000           450
    1      1002     Diya        Indian        Thane      40.000000           300
    2      1003    Kabir       Chinese       Mumbai      37.777778           550
    3      1004    Meera       Burgers  Navi Mumbai      25.000000           250
    4      1005   Ishaan        Indian         Pune      35.000000           400
    5      1006    Anaya        Indian        Thane      50.000000           350
    7      1008    Rohan       Mexican       Mumbai      20.000000           600
    8      1009     Sara         Pizza         Pune      45.000000           500
    9      1010   Vivaan       Chinese  Navi Mumbai      55.000000           700

       Rating Payment_Mode
    0     4.5          UPI
    1     4.0         Card
    2     5.0         Cash
    3     4.3          UPI
    4     4.2         Card
    5     3.8         Card
    7     4.7          UPI
    8     4.4         Cash
    9     4.1         Card

    Duplicates after removal: 0
    Dataset Shape: (9, 8)
```

Figure 6: Removal of duplicate records resulting in dataset shape reduction to (9, 8).

```
...  ===== DATASET AFTER CATEGORICAL ENCODING =====
       Order_ID Customer  Food_Category  City  Delivery_Time  Order_Amount  \
    0      1001    Aarav              4     0      30.000000           450
    1      1002     Diya              2     3      40.000000           300
    2      1003    Kabir              1     0      37.777778           550
    3      1004    Meera              0     1      25.000000           250
    4      1005   Ishaan              2     2      35.000000           400
    5      1006    Anaya              2     3      50.000000           350
    7      1008    Rohan              3     0      20.000000           600
    8      1009     Sara              4     2      45.000000           500
    9      1010   Vivaan              1     1      55.000000           700

       Rating  Payment_Mode
    0     4.5             2
    1     4.0             0
    2     5.0             1
    3     4.3             2
    4     4.2             0
    5     3.8             0
    7     4.7             2
    8     4.4             1
    9     4.1             0

    Food Category Encoding:
    {'Burgers': np.int64(0), 'Chinese': np.int64(1), 'Indian': np.int64(2), 'Mexican': np.int64(3), 'Pizza': np.int64(4)}

    City Encoding:
    {'Mumbai': np.int64(0), 'Navi Mumbai': np.int64(1), 'Pune': np.int64(2), 'Thane': np.int64(3)}

    Payment Mode Encoding:
    {'Card': np.int64(0), 'Cash': np.int64(1), 'UPI': np.int64(2)}
```

Figure 7: Categorical encoding transformation for Food_Category, City, and Payment_Mode.

```
    ===== DATASET AFTER FEATURE SCALING =====
…      Order_ID Customer  Food_Category  City  Delivery_Time  Order_Amount  \
    0      1001    Aarav              4     0      -0.697200     -0.040161
    1      1002     Diya              2     3       0.228590     -1.124507
    2      1003    Kabir              1     0       0.022859      0.682736
    3      1004    Meera              0     1      -1.160095     -1.485956
    4      1005   Ishaan              2     2      -0.234305     -0.401610
    5      1006    Anaya              2     3       1.154380     -0.763058
    7      1008    Rohan              3     0      -1.622990      1.044185
    8      1009     Sara              4     2       0.691485      0.321288
    9      1010   Vivaan              1     1       1.617275      1.767083

         Rating  Payment_Mode
    0  0.481125             2
    1 -0.962250             0
    2  1.924501             1
    3 -0.096225             2
    4 -0.384900             0
    5 -1.539601             0
    7  1.058475             2
    8  0.192450             1
    9 -0.673575             0

    ===== SCALED NUMERICAL COLUMNS =====
       Delivery_Time  Order_Amount    Rating
    0      -0.697200     -0.040161  0.481125
    1       0.228590     -1.124507 -0.962250
    2       0.022859      0.682736  1.924501
    3      -1.160095     -1.485956 -0.096225
    4      -0.234305     -0.401610 -0.384900
    5       1.154380     -0.763058 -1.539601
    7      -1.622990      1.044185  1.058475
    8       0.691485      0.321288  0.192450
    9       1.617275      1.767083 -0.673575
```

Figure 8: Standardization of continuous numerical columns (Delivery_Time, Order_Amount, Rating) using StandardScaler.

```
⋯  ===== FINAL DATASET VERIFICATION =====

    Missing Values:
    Order_ID         0
    Customer         0
    Food_Category    0
    City             0
    Delivery_Time    0
    Order_Amount     0
    Rating           0
    Payment_Mode     0
    dtype: int64

    Total Missing Values: 0

    Duplicate Rows:
    0

    Final Dataset Shape: (9, 8)

    Final Preprocessed Dataset:
       Order_ID Customer  Food_Category  City  Delivery_Time  Order_Amount  \
    0      1001    Aarav              4     0      -0.697200     -0.040161
    1      1002     Diya              2     3       0.228590     -1.124507
    2      1003    Kabir              1     0       0.022859      0.682736
    3      1004    Meera              0     1      -1.160095     -1.485956
    4      1005   Ishaan              2     2      -0.234305     -0.401610
    5      1006    Anaya              2     3       1.154380     -0.763058
    7      1008    Rohan              3     0      -1.622990      1.044185
    8      1009     Sara              4     2       0.691485      0.321288
    9      1010   Vivaan              1     1       1.617275      1.767083

         Rating  Payment_Mode
    0  0.481125             2
    1 -0.962250             0
    2  1.924501             1
    3 -0.096225             2
    4 -0.384900             0
    5 -1.539601             0
    7  1.058475             2
    8  0.192450             1
    9 -0.673575             0
```

Verification Summary

Metric / Parameter	Initial Dataset	Final Preprocessed Dataset
Total Rows	10	9
Total Columns	8	8
Total Missing Values	4	0
Duplicate Rows	1	0
Data Format	Mixed Text & Numerical	Standardized Numerical Vectors
Export Status	Raw Input Object	Saved as `Preprocessed_Food_Delivery.xlsx`

Conclusion

The Food Delivery Management dataset was successfully preprocessed using Python and the Pandas, NumPy, and Scikit-learn libraries. Missing values in continuous columns (Delivery_Time, Rating) were imputed using column means, while categorical missing values (Food_Category, Payment_Mode) were populated using column modes. Duplicate records were identified and dropped, reducing the dataset from 10 to 9 unique entries. Categorical attributes were converted to numerical representations using `LabelEncoder`, and numerical features were standardized using StandardScaler. The final dataset contains 9 rows and 8 columns with zero missing values and zero duplicate rows, and was exported to `Preprocessed_Food_Delivery.xlsx` ready for machine learning tasks.

Result

Thus, data preprocessing was successfully performed on the Food Delivery Management dataset using Python. Missing values were handled, duplicate records were removed, categorical attributes were encoded, and numerical features were standardized. The processed dataset was verified and exported in Excel format for further analysis.